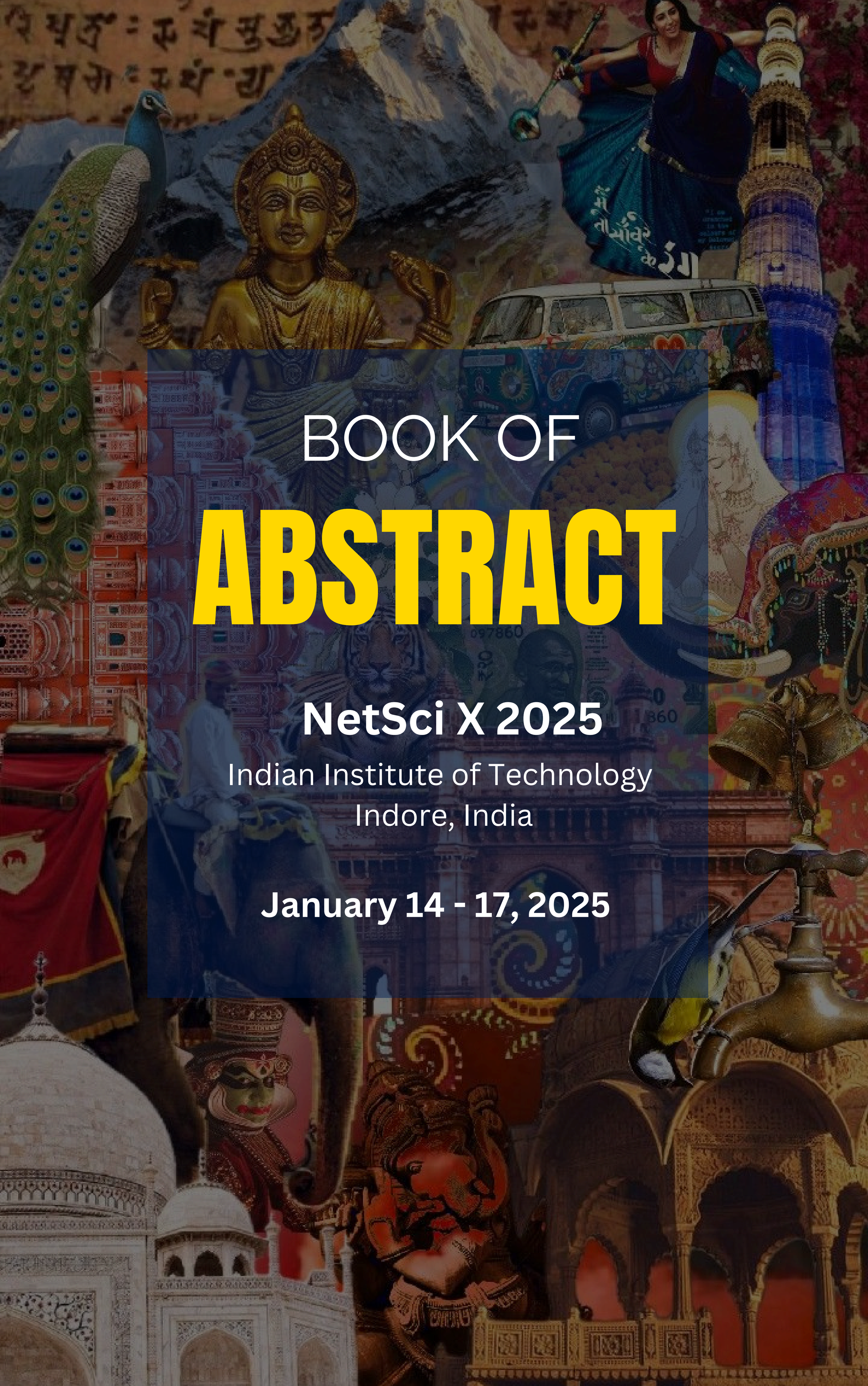

BOOK OF

# ABSTRACT

## NetSci X 2025

Indian Institute of Technology
Indore, India

**January 14 - 17, 2025**

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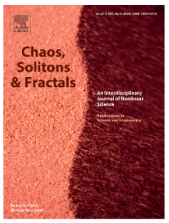


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# Rising inequality signals collapse in a network based model of complex adaptive systems

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Understanding why collapses happen in complex systems around us is significant for forecasting and response planning. In this work, we attempt to understand the origin of crashes in a mathematical model of an evolving complex network. The model [1] describes the dynamics of a directed network of $s$ nodes, represented by the adjacency matrix $A = (A_{i\ j})$, coupled to the populations $x_i$ of the nodes ($i, j = 1 \ldots s$). The network shows the spontaneous growth and collapse of structure and complexity in its time evolution.

Although previous studies have identified proximate reasons for collapses in the model [2, 3, 4], the long term dynamics that leads the network to become fragile is not understood. One of the common theoretical paradigms that study collapses is critical transitions[5], in the present work we show that collapses in this model do not fit this paradigm . We have further computed the Gini coefficient, a measure used by economists to study income inequality, for the distribution of node populations $x_i$ in the model. We find that there is an average rise in inequality among the nodes as the network approaches collapse. We show that the peripheral nodes of the network become richer at the expense of the core thereby increasing the fragility of the network to collapse. Some of the lessons from this model could be relevant for other complex adaptive systems.

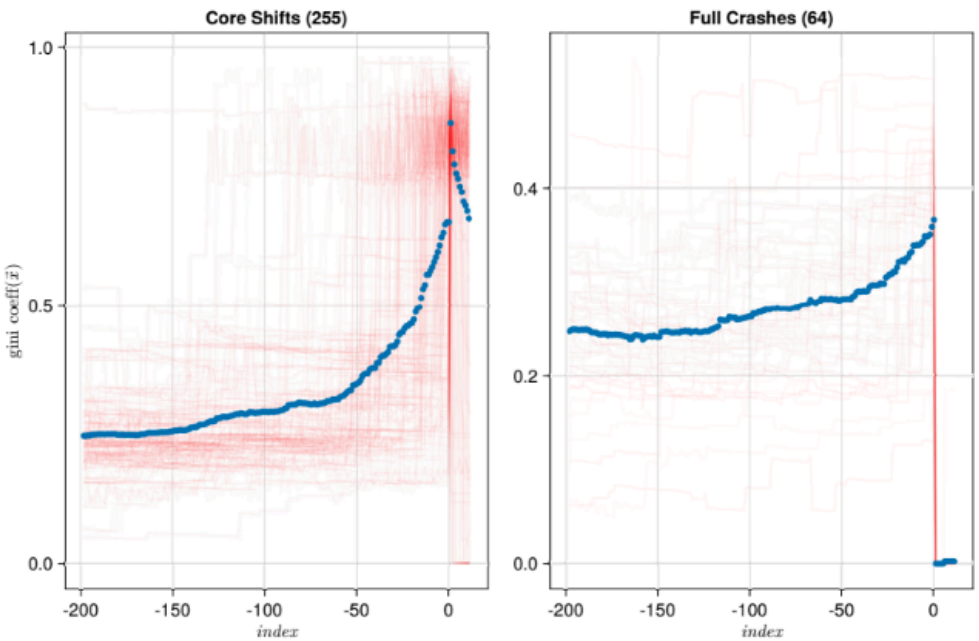


Figure 1: Rising inequality among nodes (Gini coefficient) as a function of time (collapse occurs at 0), averaged over multiple crashes. The two figures correspond to two distinct topological classes of crashes (core shifts and full crashes) in net- works of size $s = 100$.

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## A multidimensional systems biology approaches on Mtb-host interactions reveals potential regulators for host metabolic reprogramming

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Tuberculosis (TB) is a pervasive and devastating air-borne disease caused by the organisms belonging to the Mycobacterium tuberculosis (Mtb) complex. To persist and induce pathogenesis, Mtb has developed various strategies to modulate host metabolism, which is pivotal in determining various disease outcomes. But to study this through metabolomics only provides static snapshots, missing dynamics profiles of metabolite trafficking and fluxes, which are essential for understanding metabolic regulation mechanisms. Given the challenges of metabolic profiling, transcriptomic analyses serve as a proxy to uncover metabolic reprogramming and vulnerabilities. This transcriptomic data enables a systematic interrogation of gene expression patterns in biospecimens at tissue and cellular levels. Hinging on this, present study investigates the RNA-seq data of Mtb strain H37Rv infected mice lung tissue to understand host metabolic reprogramming. Preliminary analysis of RNA-seq data revealed that the differentially expressed genes are enrichment in various metabolic processes like lipid, amino acid and nucleotides metabolism. To capture the metabolic-flux profiles of infected and uninfected mice lung, we applied genome-scale metabolic model by integrating the gene expression data. The flux states analysis shows alterations in carbohydrate, amino acid, cholesterol, lipid, vitamin, nucleotide, and energy metabolism in infected mice lung. Our analysis also uncovered 8 up-regulated, and 17 down-regulated genes were responsible for these metabolic reprogramming in infected mice lungs. The same was also confirmed from two separate independent studies. Currently, we are developing network tool to link between the perturbation observed in the host metabolic flux profile capture from transcriptomic data, with various pathogen proteins.

**Keywords:** Tuberculosis; Genome-scale metabolic model; Host-pathogen interaction network; Host directed therapy

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